What Men Do with a Day Off: Sundays, Paid Work, and the Gendered Division of Unpaid Work in India

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Abstract

Do men take on unpaid work when they work fewer paid hours? Using India's Time Use Survey 2024, I instrument men's paid work with a Sunday diary, assuming that Sunday affects their unpaid work only through their paid work. For regular salaried men, a Sunday diary reduces paid work by 197 minutes (SE 6.3). Under this assumption, of every 100 minutes of paid work forgone, 16.1 (SE 0.8) go to unpaid domestic and care work, 58.3 (SE 1.1) to free time and 22.7 (SE 0.9) to sleep. Men without paid work show no Sunday change in unpaid work, supporting the exclusion restriction. The wife-minus-husband gap in unpaid work, 384 minutes on weekdays, is 34.3 minutes (SE 3.5) smaller on Sundays. Released time flows mostly into leisure and rest, which limits what shorter working hours alone can do for the gender division of unpaid work.

Keywords: weekend, working time, unpaid work, gender, instrumental variables, India
JEL codes: J22, J16, D13

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1 Introduction

Women in India do far more unpaid domestic and care work than men, and men do far more paid work (Hirway and Jose 2011; National Statistical Office 2020; National Statistics Office 2025). One explanation for the gap is time availability: men do little at home because they are at work. The explanation has a long history in the sociology of housework (Coverman 1985; Shelton and John 1996; Coltrane 2000), and it is implicit in policies that try to change the division of unpaid work by changing men's working time, such as shorter standard weeks, flexible schedules and paternity leave. If time availability is the binding constraint, then releasing men from paid work should raise their unpaid work. If it is not, the released time should go mostly to leisure and rest, and the division of unpaid work should change little.

The question is hard to answer with a cross-section. Men who work fewer hours differ from men who work more in their jobs, health, wealth, family structure and attitudes, and each of these also predicts how much unpaid work they do. Evidence from economic shocks that cut men's working time, such as recessions, job loss and the COVID-19 lockdowns, shows that some of the released time goes to home production (Aguiar et al. 2013; Gough and Killewald 2011; Sevilla and Smith 2020; Farré et al. 2022). But these shocks also cut income, raise uncertainty and, in the case of lockdowns, close schools and keep every household member at home, so they do not isolate the effect of time itself.

This paper uses a more ordinary source of variation: the day of the week. India's Time Use Survey (TUS) 2024 asks each household member aged six and over to report their activities for the 24 hours preceding the interview (National Statistics Office 2025). Interviews are spread over the week, so whether a household's diary falls on a Sunday, a Saturday or a weekday is determined by when the investigator visits, not by what the household does that day. For regular salaried men, most of whom have a fixed weekly day off, a Sunday diary removes most of a working day. The diary day is therefore a quasi-random shock to these men's paid work that leaves their income, job and household unchanged. Weekend comparisons have a long history in time-use research (Hamermesh 2002; Jenkins and Osberg 2005; Craig and Brown 2014). What the TUS design adds is that the diary day is assigned by the survey calendar rather than chosen by the respondent, which makes it possible to use the weekend as an instrument.

I use the Sunday diary as an instrument for regular salaried men's paid work and estimate how a forgone minute of paid work is reallocated across every other activity. Because every diary sums to 1,440 minutes, the instrumental-variables (IV) coefficients for the non-work activities sum to one, and each is the share of a forgone paid-work minute that goes to that activity. The sample contains 36,758 diaries of regular salaried men aged 20 to 59. A Sunday diary reduces their paid work by 197 minutes (SE 6) from a weekday mean of 526 minutes, and the share with no paid work at all is 39.8% on Sundays against 4.6% on weekdays. The first-stage F statistic is 975, far above the threshold at which conventional IV inference is reliable (Lee et al. 2022). The key identifying

assumption is that a Sunday changes these men's unpaid work only through their paid work. I test it on men aged 20 to 59 who are not in paid work, for whom Sunday changes nothing at work: their Sunday change in unpaid work is 0.1 minutes (SE 4.1), consistent with the assumption.

Four findings stand out. First, under the assumption that a Sunday affects salaried men's unpaid work only through their paid work, of every 100 minutes of paid work forgone on Sunday, 16.1 (SE 0.8) go to unpaid domestic and care work, 58.3 (SE 1.1) to free time (socialising, leisure, media and sport) and 22.7 (SE 0.9) to sleep. The unpaid-work share is positive and statistically significant, and it is smaller than the combined share that goes to free time and sleep. Second, the unpaid work that men do take on is concentrated in particular tasks: the largest share goes to shopping. Third, several tests support the exclusion restriction that Sunday affects men's unpaid work only through their paid work. Pre-determined characteristics are not fully balanced across diary days (joint $p = 0.00$; Sunday households are slightly smaller), so I condition on them and show that the estimates are unchanged with FSU fixed effects and without controls. Among men without paid work, for whom Sunday changes nothing at work, unpaid work does not change detectably on Sundays. The net estimate is 16.2 (SE 2.3) minutes per 100. Fourth, wives of regular salaried men do 1.4 minutes (SE 3.1) more unpaid work on the household's Sunday, and the wife-minus-husband gap in unpaid work is 34.3 minutes (SE 3.5) smaller on Sundays than its weekday level of 384 minutes.

The paper makes three contributions. It provides an instrumental-variables estimate, valid under an explicit exclusion assumption, of the marginal allocation of men's paid-work time in a low-income, high-gap setting, whereas the causal studies of working time and housework reviewed below are from high-income countries (Goux et al. 2014; Kawaguchi et al. 2013; Patnaik 2019). It shows how a common feature of time-use survey design, the "yesterday" diary combined with interviews spread over the week, can be used for identification, with a set of falsification tests that other surveys with the same design can reproduce. And it links the husband's response to the wife's response within the same couple on the same day, which speaks directly to whether shorter working time redistributes unpaid work or only adds leisure.

The rest of the paper proceeds as follows. Section 2 reviews the literature. Section 3 sets out a simple framework. Section 4 describes the data and Section 5 the design. Section 6 presents the results, Section 7 discusses interpretation and policy, and Section 8 concludes.

2 Related literature

2.1 Total work, time availability and the division of unpaid work

The economic analysis of time use begins with the household as a producer that combines market goods and members' time (Becker 1965), with specialisation by comparative advantage (Becker 1991) and, in later models, with bargaining between spouses whose outside options shape the allocation (Lundberg and Pollak 1993). In these models a fall in one spouse's market hours, holding wages fixed, should raise that spouse's home production, by an amount that depends on how substitutable

spouses' home time is and on how much the released time is valued as leisure.

Descriptive work on total work, the sum of paid and unpaid work, finds that men and women in rich countries do similar amounts of total work, while in poorer countries women's total work tends to exceed men's (Burda et al. 2013). Over the long run, women's paid work has risen and their unpaid work has fallen, while men's unpaid work has risen slowly (Gershuny 2000; Bianchi et al. 2000; Sayer 2005; Gimenez-Nadal and Sevilla 2012; Aguiar and Hurst 2007). Convergence has been uneven across countries and tasks. Men's participation has risen more in some institutional settings than in others (Gershuny and Sullivan 2003; Hook 2006; Kan et al. 2011), and the segregation of housework into "male" and "female" tasks has persisted (Hook 2010).

Sociologists distinguish three explanations for the division of housework: time availability, relative resources and gender ideology (Coverman 1985; Shelton and John 1996; Coltrane 2000). Time availability predicts that each spouse's housework falls with their own paid hours. Relative resources predicts that the spouse with more earnings or bargaining power does less. Gender ideology and identity predict that norms about what men and women should do constrain the division regardless of time or money. The evidence that husbands who are economically dependent on their wives do less, not more, housework (Brines 1994), that women's housework declines with their relative earnings only up to a point (Bittman et al. 2003), and that couples in which the wife out-earns the husband show patterns consistent with gender identity (Akerlof and Kranton 2000; Bertrand et al. 2015) all suggest that time availability is not the whole story. These studies, however, rely on cross-sectional or panel variation in hours and earnings that is itself chosen. The design here holds income, job and household fixed and moves only time.

2.2 What people do with released time

A body of work estimates how time released from paid work is reallocated. Aguiar et al. (2013) use the American Time Use Survey over the Great Recession and find that roughly 30 percent of forgone market work hours were reallocated to home production and about half to leisure, much of it sleep and television. Burda and Hamermesh (2010) compare the time use of the unemployed and the employed across countries, and Gough and Killewald (2011) study how men's and women's housework changes with unemployment within couples. Retirement offers a further, predictable reduction in working time, and Stancanelli and Van Soest (2012) use a regression discontinuity at retirement age to estimate its effect on home production. Each of these shocks changes more than time: unemployment and recessions cut income and raise uncertainty, and retirement is permanent and anticipated.

Working-time regulation is closer to the experiment of interest. Goux et al. (2014) study the French reduction of the standard work week to 35 hours and find that the reduction in husbands' hours also reduced their wives' hours, consistent with complementarity in spouses' leisure. Kawaguchi et al. (2013) study how Japanese and Korean workers reallocated time when statutory standard hours were cut, and Hamermesh et al. (2017) examine workers' well-being after the Korean reduction. In

a closely related use of day-to-day variation, Connolly (2008) shows that men in the United States substitute between work and leisure across rainy and dry days. The present paper uses a different, recurring and plausibly exogenous source of variation, the weekly day off, in a country where men's baseline unpaid work is very low.

2.3 Paternity leave and the COVID-19 lockdowns

Two other sources of released male time have been studied. Father quotas in parental leave raise fathers' later involvement in household production and childcare in some settings (Patnaik 2019; Tamm 2019) but not in others (Ekberg et al. 2013); leave, however, combines released time with an explicit expectation of care. The 2020 lockdowns raised men's childcare and housework modestly while women continued to do most of it, in the United Kingdom (Sevilla and Smith 2020), Spain (Farré et al. 2022) and India (Deshpande 2020), but they also closed schools and removed paid help, so they mix released time with a large rise in the unpaid work to be done. A day off comes closer to isolating the time channel, although it also changes who else is at home.

2.4 Timing, synchronisation and the weekend

A separate literature studies when, rather than how much, people work. Spouses value time together and coordinate the timing of their work and leisure (Hamermesh 2002; Hallberg 2003). The value of leisure depends on whether others are free at the same time (Jenkins and Osberg 2005), and spouses' non-market time choices depend on each other's (Connelly and Kimmel 2009). Non-standard schedules reduce the time couples and families spend together (Presser 2003; Lesnard 2008). When dual-earner spouses work different shifts, husbands do more of the tasks traditionally done by women (Presser 1994). Craig and Powell (2011) study how non-standard schedules relate to the division of childcare, and Craig and Brown (2014) show that weekend work reduces leisure time with family and friends, with different costs for men and women. Bryan and Sevilla (2017) examine whether flexible working helps couples coordinate their time, and Hamermesh and Stancanelli (2015) document that Americans work at nights and on weekends more often than Europeans.

The weekend therefore has two distinct roles. It releases time, and it synchronises: when the husband's day off coincides with children's school holiday and with other households' days off, the household is together, and the released time may be used jointly rather than in unpaid work. This makes the weekend an informative experiment for policy, because shorter working weeks would usually be implemented as longer weekends.

2.5 Men's unpaid work in India

India's gender gap in unpaid work is large and persistent. It is documented in the 1998–99 pilot time-use survey (Hirway and Jose 2011), the TUS 2019 (National Statistical Office 2020) and the TUS 2024 (National Statistics Office 2025). The gap coexists with low and stagnant female labour force participation (Klasen and Pieters 2015; Afridi et al. 2018; Chatterjee et al. 2018;

Fletcher et al. 2017). Social norms about women's work outside the home are a central explanation (Jayachandran 2021), and interventions that change women's control over earnings can shift both labour supply and norms (Field et al. 2021). Norms about men's unpaid work have received less attention. Experimental evidence from other settings shows that men can misperceive the norms held by their peers (Bursztyn et al. 2020), which suggests that men's unpaid work may respond to the visibility of other men's behaviour as well as to time. This paper asks the prior question: whether, when the constraint of paid work is lifted for a day, Indian men use the time for unpaid work at all.

3 A simple framework

Let a man's day of $T = 1{,}440$ minutes be divided into paid work h and K other activities $a_1, \ldots, a_K$, so that $h + \sum_k a_k = T$. On a working day, h is set largely by the employer. On a day off, h falls by $\Delta h < 0$ for reasons unrelated to the man's preferences that day. The time constraint implies

$$\sum_{k=1}^{K} \frac{\Delta a_k}{-\Delta h} = 1.$$

Each term $s_k = \Delta a_k/(-\Delta h)$ is the share of a forgone paid-work minute allocated to activity k. Time availability predicts a large share for unpaid work s_u. A model in which men's unpaid work is fixed by norms or by the division of tasks within the household predicts $s_u \approx 0$, with the released time going to leisure and sleep.

The wife's response is informative about how spouses' time interacts. If spouses' unpaid work is substitutable, the wife's unpaid work falls when the husband's rises, and the within-couple gap narrows by more than the husband's increase. If spouses value joint leisure (Hamermesh 2002; Goux et al. 2014), the wife's free time rises on the husband's day off and her unpaid work may fall even when his does not rise. If the household's demand for unpaid work rises on Sundays, because children are at home or because weekly tasks are done on the day off, both spouses' unpaid work rises. The data distinguish these cases by the joint sign pattern of the wife's unpaid work, her free time and the husband's unpaid work.

Two features of a day off differ from a permanent reduction in working time. A day off is anticipated and recurs weekly, so tasks that can be postponed, such as shopping or repairs, may be moved to it from other days. This intertemporal reallocation would make the Sunday share of unpaid work larger than the share under a permanent reduction. A day off is also a common holiday for children and many other workers. This synchronisation would make leisure more valuable on Sundays than on a day off taken alone, and the Sunday share of leisure larger than under a staggered reduction. Section 7 returns to these points.

4 Data

4.1 The Time Use Survey 2024

The TUS 2024 was conducted by the National Statistics Office from January to December 2024 (National Statistics Office 2025). It covers a nationally representative sample of households, and every member aged six and over reports their activities in 30-minute slots for the 24 hours from 4 a.m. on the day before the interview. Activities are coded to the International Classification of Activities for Time-Use Statistics 2016 (United Nations Statistics Division 2016). The analysis uses every activity record (10,211,478 records, 450,457 diaries in 139,487 households), aggregated to one row per person-day. Minutes are computed from major activities, so every diary sums to 1,440 minutes (Appendix Table 11). All estimates use the survey weights and cluster standard errors on the first-stage unit (FSU), the primary sampling unit (Abadie et al. 2023; Cameron and Miller 2015; Solon et al. 2015).

The outcome categories follow the ICATUS divisions: employment and related activities (paid work, including work-related travel and commuting); production of goods for own final use; unpaid domestic services; unpaid caregiving for household members; volunteer and other unpaid work; learning; socialising, community participation and religious practice; culture, leisure, mass media and sport; and self-care and maintenance, which includes sleep. "Unpaid work" is the sum of unpaid domestic services and unpaid care, the SDG 5.4.1 definition. "Free time" is the sum of socialising and leisure. Unpaid domestic work is split into its ICATUS sub-divisions to study which tasks men take on.

4.2 Samples

The main sample is men aged 20 to 59 whose usual principal activity status is regular wage or salaried employment, who make up 27.0% of employed men in this age range. They are the group for whom the weekly day off is most clearly set by the employer. Indian labour law requires a weekly holiday for factory workers, by default on the first day of the week (Factories Act, 1948, Section 52). Figure 1 shows how paid work varies by diary day in the data. Placebo samples are men of the same age who are casual labourers, self-employed, students, or not in paid work (unemployed, attending to domestic duties, or otherwise out of the labour force). The couple sample links wives aged 20 to 59 to their husbands through the household roster (head and spouse, or married child and child's spouse). In 100.0% of linked couples both spouses' diaries refer to the same day; I keep these couples, of whom 24,031 have a regular salaried husband.

Table 1 shows the raw contrast. Regular salaried men work 526 paid minutes on an average weekday and do 41 minutes of unpaid work; their wives do 426 minutes of unpaid work. The columns for Sunday show how both spouses' days change when the husband's working day is removed.

Table 1: Minutes per day by diary day: regular salaried men 20–59 and their wives

Outcome	Men Mon–Fri	Men Sat	Men Sun	Wives Mon–Fri	Wives Sat	Wives Sun
Paid work	526	510	332	85	79	42
Unpaid domestic services	23	27	50	347	347	354
Unpaid care	18	19	24	79	82	77
Free time	187	195	298	225	225	245
Sleep	494	498	539	526	529	544
Any unpaid work (share)	0.45	0.47	0.56	0.99	0.99	0.99
Diaries	26,874	5,230	4,654	17,758	3,366	2,907

Notes: Survey-weighted means. Wives are linked spouses of regular salaried men whose diaries refer to the same day as their husband's. Free time is socialising, community and religious activities plus culture, leisure, mass media and sport.

5 Empirical strategy

5.1 Assignment of the diary day

The diary refers to the day before the interview, and the interview date is set by the investigator's schedule within the FSU. If that schedule is unrelated to households' characteristics, conditional on location and survey period, the diary day is as good as randomly assigned. Two features of the data make this plausible. Diary days are spread evenly over the week (Appendix Table 10): the weighted share of diaries on each day lies between 13.7% and 14.5%. The diary day also varies within FSUs: in 99.9% of FSUs, sampled households report on more than one day. Within households, 100.0% of households have all members' diaries on the same day, so the diary day is in practice a household-level assignment.

The main threat to assignment is that investigators might find some households at home only on particular days. Households in which the man works on weekdays might be interviewed disproportionately on Mondays (diary day Sunday) or on Sundays (diary day Saturday), for example. Table 2 tests this by regressing pre-determined characteristics on Sunday and Saturday indicators, with state and survey sub-round fixed effects, and reports a joint test of all six day-of-week indicators for each characteristic, and a joint test of all characteristics in a regression of the Sunday indicator on them.

5.2 Instrumental variables

For regular salaried men I estimate

$$y_i = \theta\, h_i + \beta\, Sat_i + X_i'\gamma + \mu_s + u_i, \qquad h_i = \pi\, Sun_i + \rho\, Sat_i + X_i'\delta + \mu_s + v_i,$$

where h_i is minutes of paid work, y_i is minutes in another activity, Sun_i and Sat_i indicate the diary day, X_i contains age band, education, marital status, urban residence, consumption quintile, household size and the numbers of household members under 6, aged 6 to 14 and aged 60 or over,

and μ_s are state and sub-round fixed effects. Saturday enters as a control, so the instrument contrasts Sunday with Monday to Friday. The share of a forgone paid-work minute allocated to activity y is $s_y = -\theta$. Because the outcomes partition the day, the estimated shares across the eight non-employment ICATUS divisions sum to exactly one (here, 1.00).

Under monotonicity, the IV estimate is a local average effect for compliers (Imbens and Angrist 1994; Angrist and Pischke 2009): regular salaried men who work on weekdays and not on Sundays. Men who work every day, and men who would not have worked that weekday anyway, do not contribute. This is the population of interest for a policy that shortens the standard working week of salaried employees.

I report the first-stage Wald F statistic computed with the clustered variance, which equals the effective F of Montiel Olea and Pflueger (2013) with a single instrument. When it exceeds 104.7, the conventional 5% t-test has correct size without adjustment (Lee et al. 2022).

The outcomes contain many zeros, since many men do no unpaid work on a given day. I estimate linear models in minutes, which recover population means of time use without the distributional assumptions of censored models (Stewart 2013). Single-day diaries measure time on a day, not a person's typical day (Frazis and Stewart 2012). Here this is the point: the estimand is the response of a day's time use to that day's paid work.

5.3 The exclusion restriction

The exclusion restriction requires that Sunday affects a regular salaried man's unpaid work only through his paid work. Four channels could violate it. (i) Other household members are also at home on Sundays: children are out of school and wives who work are off. Their presence changes the amount of unpaid work to be done and who is available to do it. (ii) Markets and services operate differently on Sundays: some shops close and weekly markets meet on fixed days. (iii) Religious and social activities are concentrated on particular days. (iv) Reporting differs by day: a Sunday diary is reported on a Monday, when the man is more likely to be at work and to have his diary reported by another household member. These channels mostly operate through the household's day, not the man's job, so they should also be present for men whose paid work does not change on Sundays.

I test this in three ways. The falsification test estimates the Sunday effect on unpaid work among men aged 20 to 59 who are not in paid work. For them, Sunday changes nothing at work, so any Sunday effect on their unpaid work is a direct effect of the day. Students provide a second placebo group, and the Sunday effect on children's learning time measures how much the school calendar changes the household's day. Second, a difference-in-reduced-forms IV pools regular salaried men with men not in paid work and uses the interaction of Sunday and regular salaried status as the instrument, controlling for Sunday and Saturday main effects and for status. This removes any direct Sunday effect that is common to the two groups. Third, I report how the estimated unpaid-work share changes under an assumed direct Sunday effect of γ minutes, by re-estimating the IV with

$y_i - \gamma\, Sun_i$ as the outcome (Conley et al. 2012).

Casual labourers and the self-employed are not clean placebos: Sunday can reduce their paid work too, but less predictably. Their first stages show how specific the weekly day off is to salaried employment. The Saturday indicator serves as a second instrument. Many salaried men work a full or half day on Saturdays, so Saturday is a weaker but distinct shock, and the comparison of Saturday and Sunday IV estimates tests whether the allocation of released time depends on which day is released.

6 Results

6.1 Balance

Table 2: Balance of pre-determined characteristics by diary day

Characteristic	Sunday	SE	Saturday	SE	p, all days
Age	0.00	0.06	-0.13**	0.06	0.06
Female	-0.00**	0.00	-0.00	0.00	0.14
Married	-0.01***	0.00	-0.01***	0.00	0.00
Urban	-0.00	0.00	-0.00*	0.00	0.31
Household size	-0.22***	0.02	-0.07***	0.02	0.00
Members under 6	-0.00	0.01	-0.01	0.01	0.18
Members aged 6–14	-0.13***	0.01	-0.03***	0.01	0.00
Members aged 60+	-0.01	0.01	-0.00	0.01	0.42
Graduate or above	-0.01***	0.00	-0.00	0.00	0.01
Log monthly per-capita expenditure	0.01***	0.00	0.00	0.00	0.04
Scheduled Caste or Tribe	-0.00	0.00	0.01	0.00	0.54
Muslim	-0.00	0.00	0.00	0.00	0.94
Employed (usual status)	-0.01**	0.00	-0.00	0.00	0.08
Regular salaried (usual status)	-0.02***	0.00	-0.00	0.00	0.00
Joint test, all characteristics (p): all persons	0.00		0.12		
Joint test, all characteristics (p): regular salaried men	0.00				

Notes: Persons 20–59 (293,338 diaries). Each row regresses the characteristic on Sunday and Saturday indicators (Monday to Friday omitted) with state and sub-round fixed effects. 'p, all days' is the p-value of a joint Wald test that all six day-of-week indicators are zero. The joint tests regress the Sunday (Saturday) indicator on all listed characteristics and test that their coefficients are jointly zero. FSU-clustered SEs. * p<0.10, ** p<0.05, *** p<0.01.

Table 2 supports quasi-random assignment only in part. Of 14 characteristics, 8 differ between Sunday and weekday diaries at the 5% level, against 0.7 expected by chance. The joint test that all characteristics are unrelated to a Sunday diary has p = 0.00 for all persons and p = 0.00 among regular salaried men. At least one rejects at the 5% level, so the controls and the robustness checks with FSU fixed effects carry more weight than they would under clean assignment. All estimates condition on these characteristics.

The design does predict one feature that is not pre-determined: who reports the diary. A Sunday diary is collected on a Monday, when salaried men are more likely to be at work. Among regular salaried men, a diary is 4.6 percentage points (SE 0.9) less likely to have been reported by another

household member when it refers to a Sunday, and 2.8 percentage points (SE 0.8) less likely when it refers to a Saturday, which is collected on a Sunday. Proxy reporting is therefore an outcome of the design rather than a control. I exclude it from the main specification and show results with it as a control and for self-reported diaries only.

6.2 Paid work by diary day

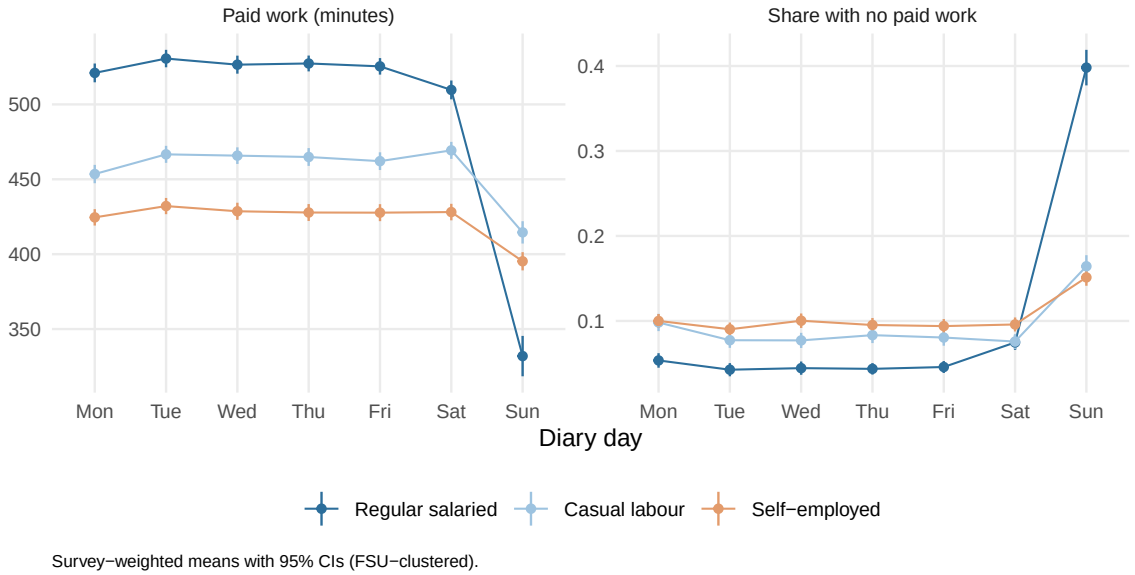


Figure 1: Paid work by diary day: men 20–59 by usual activity status

Figure 1 plots mean paid work and the share of men with no paid work by diary day. For regular salaried men, paid work is flat from Monday to Friday and falls sharply on Sunday; the share doing no paid work rises from 4.6% on weekdays to 39.8% on Sundays. Casual labourers' paid work changes by -48.2 minutes (SE 3.8) on Sundays and self-employed men's by -31.1 minutes (SE 2.9). Both are smaller than the change for salaried men, consistent with the weekly day off being a feature of salaried employment.

6.3 How men reallocate released time

Table 3 reports the reduced form and the IV allocation. Of every 100 minutes of paid work forgone on Sunday, 16.1 (SE 0.8) go to unpaid domestic and care work, 58.3 (SE 1.1) to free time and 22.7 (SE 0.9) to sleep. The remainder goes to own-use production, volunteering, learning and self-care other than sleep. In minutes, a Sunday changes these men's unpaid work by 31.8 minutes (SE 1.7), their free time by 115.0 minutes (SE 4.2) and their sleep by 44.7 minutes (SE 2.4). The unpaid-work share is positive and statistically significant, and its 95% confidence interval, [14.5, 17.7] minutes per 100, excludes zero. Less than a quarter of the released time goes to unpaid work, a smaller share than the roughly 30 percent of forgone market hours that went to home production in the

Table 3: Allocation of paid-work time forgone on Sunday: regular salaried men 20–59

Activity	Weekday mean	Sunday effect (min)	SE	Share of forgone minute	SE
Paid work (first stage)	526	-197.3***	6.3		
Own-use production of goods	4	2.6***	0.6	0.013***	0.003
Unpaid domestic services	23	26.2***	1.5	0.133***	0.007
Unpaid care for household members	18	5.6***	0.9	0.029***	0.004
Volunteer and other unpaid work	1	0.8***	0.3	0.004***	0.002
Learning	2	-0.0	0.4	-0.000	0.002
Socialising, community, religion	66	47.1***	2.5	0.239***	0.010
Culture, leisure, mass media, sport	121	67.9***	2.8	0.344***	0.010
Self-care and maintenance	680	47.0***	2.6	0.238***	0.009
of which: sleep	494	44.7***	2.4	0.227***	0.009
Memo: all unpaid domestic and care	41	31.8***	1.7	0.161***	0.008
Memo: free time (socialising + leisure)	187	115.0***	4.2	0.583***	0.011

Notes: N = 36,758. 'Sunday effect' is the reduced-form coefficient on a Sunday diary (Saturday, controls and fixed effects included). 'Share of forgone minute' is minus the 2SLS coefficient on paid work, instrumented by the Sunday diary. The shares of the eight ICATUS divisions sum to 1.00. First-stage F = 975. FSU-clustered SEs. * p<0.10, ** p<0.05, *** p<0.01.

United States during the Great Recession (Aguiar et al. 2013), although the two shocks differ in duration and in their effect on income.

The released time is used mainly for rest and leisure: the combined share of free time and sleep is 81.0 minutes per 100, larger than the unpaid-work share.

6.4 Which unpaid tasks

Table 4: Unpaid tasks taken on when paid work is released: regular salaried men 20–59

Task	Weekday mean	Sunday effect (min)	SE	Per 100 min forgone	SE
Food and meals	8.8	1.8***	0.7	0.9	0.3
Cleaning and upkeep	2.0	2.1***	0.3	1.1	0.1
Care of textiles	1.6	2.3***	0.4	1.2	0.2
Shopping	6.8	12.3***	0.8	6.2	0.4
Repair and maintenance	0.7	1.3***	0.2	0.6	0.1
Household management	0.4	0.3*	0.2	0.2	0.1
Pets, domestic travel, other	2.9	6.0***	0.4	3.1	0.2
Childcare	16.6	5.7***	0.8	2.9	0.4
Care of adults and other care	1.4	-0.1	0.3	-0.0	0.1

Notes: Reduced-form Sunday coefficients and 2SLS shares (minus the coefficient on paid work instrumented by Sunday, times 100) for ICATUS sub-divisions of unpaid domestic services and unpaid care. '–' marks tasks with no variation in the sample. Controls and fixed effects as in Table 3. FSU-clustered SEs. * p<0.10, ** p<0.05, *** p<0.01.

Table 4 splits unpaid work into tasks. The largest share of the released time that goes to unpaid work is spent on shopping (6.2 minutes per 100). Routine, traditionally female tasks (food preparation, cleaning and the care of textiles) together absorb 3.2 minutes per 100, against 7.0 for shopping, repairs and household management, tasks that are more often shared or done by men (Hook 2010). The response is thus concentrated in tasks that can be batched into a weekly day off and that do

not displace the wife's daily routine. The share of men doing any unpaid work, 44.9% on weekdays, is 10.6 percentage points (SE 0.9) higher on Sundays, so part of the response is on the extensive margin.

6.5 Saturday as a separate shock

Table 5: Alternative instruments and the falsification-based estimator

Specification	Unpaid work per 100 min	SE	Free time per 100 min	SE	First-stage F
Sunday instrument (baseline)	16.1	0.8	58.3	1.1	975
Saturday instrument	25.4	6.8	51.2	8.5	33.1
Sunday and Saturday instruments	16.1	0.8	58.3	1.1	488
Difference-in-reduced-forms (regular vs. not in paid work)	16.2	2.3	52.5	3.4	677

Notes: Regular salaried men 20–59 (last row adds men 20–59 not in paid work, 7,765 diaries). Row 1 instruments paid work with Sunday, controlling for Saturday; row 2 with Saturday, controlling for Sunday; row 3 with both (Sargan overidentification p = 1.00). Row 4 instruments paid work with Sunday x regular salaried, controlling for Sunday, Saturday, Saturday x regular salaried and status. FSU-clustered SEs.

A Saturday diary changes regular salaried men's paid work by -17.8 minutes (SE 3.1), less than a Sunday diary. Saturday is therefore a partial day off for many of these men. Table 5 compares the allocation implied by each day. The Saturday instrument implies 25.4 (SE 6.8) minutes of unpaid work per 100 forgone, against 16.1 for Sunday. The Sargan overidentification test of equal allocations has p = 1.00; it assumes homoskedastic errors and is indicative only. It does not reject, which is consistent with the allocation of released time not depending strongly on which day is released.

6.6 Placebo groups and falsification

Table 6: Sunday effects by usual activity status: men 20–59 (minutes)

Group	Paid work	Unpaid work	Free time	Learning	Diaries
Regular salaried	-197.3*** (6.3)	31.8*** (1.7)	115.0*** (4.2)	-0.0 (0.4)	36,758
Casual labour	-48.2*** (3.8)	8.6*** (1.3)	29.4*** (2.8)	0.1 (0.2)	33,470
Self-employed	-31.1*** (2.9)	3.5*** (1.0)	22.2*** (2.2)	-0.0 (0.2)	55,734
Students	3.0 (3.5)	16.8*** (3.6)	152.7*** (8.0)	-224.2*** (8.6)	6,106
Not in paid work	-2.3 (3.8)	0.1 (4.1)	14.3** (6.5)	-14.8*** (5.1)	7,765

Notes: Reduced-form coefficients on a Sunday diary (SE in parentheses), with Saturday, controls and fixed effects. Not in paid work: unemployed, attending to domestic duties, or otherwise out of the labour force (excluding students). '–' marks outcomes with no variation in the group. FSU-clustered SEs. * p<0.10, ** p<0.05, *** p<0.01.

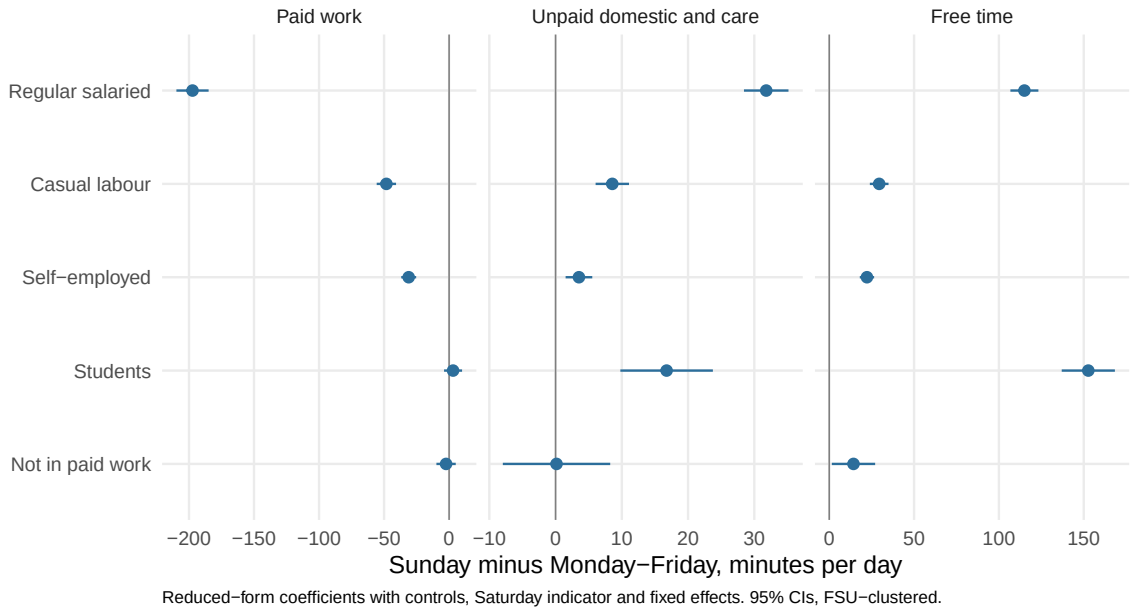


Figure 2: Sunday effects by usual activity status, men 20–59

Table 6 and Figure 2 report the reduced-form Sunday effects for each group of men. The falsification test is the row for men not in paid work. Their paid work changes by -2.3 minutes (SE 3.8) on Sundays, and their unpaid work changes by 0.1 minutes (SE 4.1). The latter is not statistically distinguishable from zero, so there is no evidence of a direct Sunday effect on men's unpaid work that operates outside paid work. Their free time changes by 14.3 minutes (SE 6.5), so Sunday raises free time even for men who are not released from work, a sign that the day's social calendar matters for leisure.

The school calendar is a channel through which Sunday could change the household's demand for unpaid work. Children aged 6 to 14 who are students spend 416 minutes on learning on an average weekday, and on Sundays they do 258.8 minutes (SE 3.0) less learning. Children are therefore at home for much of the day on Sundays. Heterogeneity by the presence of school-age children (below) speaks to whether this matters for fathers' response.

The difference-in-reduced-forms IV in Table 5 nets out any direct Sunday effect common to salaried men and men not in paid work. It implies 16.2 (SE 2.3) minutes of unpaid work per 100 forgone, within the confidence interval of the baseline estimate. The identifying assumption is weaker than the exclusion restriction but not innocuous: it requires that the direct Sunday effect on unpaid work be the same for men with and without a salaried job. Men not in paid work are a small and selected group, and the estimate is correspondingly less precise.

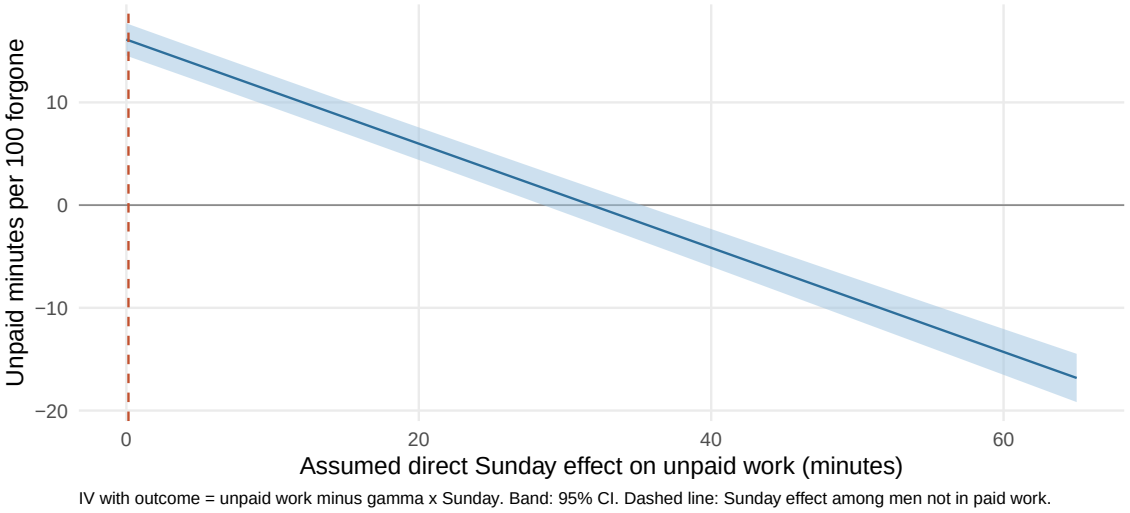


Figure 3: Unpaid-work share under an assumed direct Sunday effect on unpaid work

Figure 3 shows how the unpaid-work share changes if Sunday raised these men's unpaid work directly by γ minutes. The share falls to zero at a direct effect of 31.8 minutes, the full reduced-form effect; any positive direct effect lowers it. The dashed line marks the Sunday effect among men not in paid work, a natural benchmark for γ.

6.7 Heterogeneity

Table 7: Heterogeneity in the allocation of released time: regular salaried men 20–59

Subgroup	First stage (min)	Unpaid per 100 min	Free time per 100 min	p, difference	Diaries
Child under 6 in household: Yes	-189 (9.3)	19.6 (1.3)	56.0 (1.8)	0.00	13,459
Child under 6 in household: No	-202 (8.2)	14.2 (1.0)	59.5 (1.3)		23,299
Child aged 6–14 in household: Yes	-200 (12.8)	19.5 (1.7)	55.5 (2.4)	0.13	9,210
Child aged 6–14 in household: No	-197 (6.8)	15.3 (0.9)	59.0 (1.2)		27,548
Urban: Yes	-226 (8.8)	15.4 (1.0)	58.5 (1.2)	0.38	23,069
Urban: No	-157 (8.7)	17.4 (1.4)	57.7 (2.1)		13,689
Wife employed (usual status): Yes	-200 (16.4)	14.8 (2.0)	60.1 (3.1)	0.13	5,194
Wife employed (usual status): No	-203 (8.2)	18.5 (1.0)	56.6 (1.3)		18,510

Notes: IV estimates within subgroups (SE in parentheses). 'p, difference' tests equality of the unpaid-work share across the two subgroups in a pooled model that interacts paid work, the Sunday instrument, the Saturday control and a subgroup indicator. Wife employed is defined for married men with a linked spouse. FSU-clustered SEs.

Table 7 reports the allocation separately by household and location. Time availability predicts the largest unpaid-work response where the demand for unpaid work is largest, in households with young children. Among regular salaried men with a child under six in the household, the unpaid-work share is 19.6 (SE 1.3) minutes per 100, against 14.2 (SE 1.0) without. With a school-age child it is 19.5 (SE 1.7), against 15.3 (SE 0.9). In urban areas it is 15.4 (SE 1.0), against 17.4 (SE 1.4) in rural areas. The difference is statistically significant at the 5% level for: child under 6 in household.

Table 8: The household's Sunday: wives of regular salaried men and couple outcomes

Outcome	Weekday mean	Sunday effect	SE
Wife: paid work	85	-37.0***	2.9
Wife: unpaid domestic and care	426	1.4	3.1
Wife: childcare	76	-3.2*	1.8
Wife: free time	225	19.6***	3.3
Wife: sleep	526	15.3***	2.2
Husband: unpaid domestic and care	42	35.6***	2.1
Couple: total unpaid work	468	37.0***	4.0
Couple: wife minus husband unpaid work	384	-34.3***	3.5
Couple: husband's share of unpaid work	0.08	0.05***	0.00
Couple: upper bound on joint free time	145	46.8***	3.3
Couple: total paid work	608	-238.3***	8.4

Notes: Couples in which the husband is regular salaried and both diaries refer to the same day (24,031 couples). Reduced-form coefficients on a Sunday diary with Saturday, the wife's controls (including her usual employment status) and fixed effects. Minutes except the husband's share (0 to 1; couples with some unpaid work). The upper bound on joint free time is the smaller of the two spouses' free time; the survey's episode timing would be needed for exact overlap. FSU-clustered SEs. * p<0.10, ** p<0.05, *** p<0.01.

6.8 Wives and couples

Table 8 turns to the couple. On the household's Sunday, wives of regular salaried men do 1.4 minutes (SE 3.1) more unpaid work, 19.6 minutes (SE 3.3) more free time and 37.0 minutes (SE 2.9) less paid work than on a weekday. Wives' unpaid work does not change detectably, so the husband's day off does not relieve the wife of unpaid work. Scaled by the husband's first stage, the wife's unpaid work changes by 0.7 (SE 1.6) minutes per 100 minutes of the husband's paid work forgone, and the couple's total unpaid work by 18.4 (SE 2.0).

The wife-minus-husband gap in unpaid work, 384 minutes on an average weekday, is 34.3 minutes (SE 3.5) smaller on Sundays. The husband's share of the couple's unpaid work, 0.08 on weekdays, is 5.2 percentage points (SE 0.3) higher on Sundays. The division of unpaid work is more equal on Sundays, although the gap remains large.

Joint free time is the time both spouses are free at once. Without the timing of each spouse's episodes, I bound it from above by the smaller of the two spouses' free time. On weekdays this upper bound is 145 minutes, and it changes by 46.8 minutes (SE 3.3) on Sundays. The household's Sunday raises the free time spouses can spend together, consistent with the value of synchronised leisure (Hamermesh 2002; Jenkins and Osberg 2005).

The wife's response mixes the husband's day off with the household's Sunday: her own working day, the children's school day and the weekly calendar also change. Comparing wives of salaried men with wives of men not in paid work isolates the part due to the husband's day off. The difference in their Sunday effects on unpaid work is -7.3 minutes (SE 7.1), and on free time 14.4 minutes (SE 6.7). The husband's day off itself does not detectably change the wife's unpaid work beyond the

household's Sunday.

6.9 Robustness

Table 9: Robustness of the allocation estimates: regular salaried men 20–59

Specification	First stage	F	Unpaid per 100 min	Free time per 100 min	N
Baseline	-197 (6.3)	975	16.1 (0.8)	58.3 (1.1)	36,758
No controls (fixed effects only)	-194 (6.6)	859	17.0 (0.9)	57.5 (1.1)	36,758
Adds proxy-report indicator	-196 (6.3)	962	15.9 (0.8)	58.4 (1.1)	36,758
Self-reported diaries only	-200 (8.2)	591	16.3 (1.1)	57.0 (1.4)	19,786
FSU fixed effects	-203 (6.7)	930	16.5 (0.8)	58.5 (1.1)	35,196
Sunday against Monday–Friday only	-197 (6.3)	977	16.1 (0.8)	58.3 (1.1)	31,528
Married men only	-198 (7.0)	809	17.6 (0.9)	57.3 (1.2)	27,212
Unweighted	-203 (4.6)	1,920	16.7 (0.7)	59.0 (0.9)	36,758

Notes: IV estimates (SE in parentheses) of minutes of unpaid work and free time per 100 minutes of paid work forgone, instrumented by a Sunday diary. FSU fixed effects compare diaries within the same village or urban block; FSUs with a single diary drop out. FSU-clustered SEs.

Table 9 varies the specification. The checks drop all controls, which should matter little under random assignment; replace state fixed effects with FSU fixed effects, which compare households in the same village or urban block interviewed on different days and absorb local differences in interview schedules; control for proxy reporting or keep only self-reported diaries, because diaries reported by another household member may understate men's unpaid work (Lee and Waite 2005); drop Saturday diaries, so that Sunday is compared only with Monday to Friday; restrict to married men; and drop the survey weights (Solon et al. 2015). Across the 8 specifications the unpaid-work share ranges from 15.9 to 17.6 minutes per 100. Every estimate lies within the 95% confidence interval of the baseline, so the conclusions do not depend on these choices.

7 Discussion

7.1 What the Sunday estimate measures

The estimate is the share of a forgone paid-work minute that regular salaried men allocate to unpaid work on their weekly day off. Three features of the experiment shape how it generalises.

First, the day off is anticipated and recurs. Weekly tasks such as shopping, repairs and dealing with officials can be moved to it, which would make the unpaid-work share larger on Sundays than under a permanent reduction spread over the week. The task breakdown fits this: much of the response is in shopping, repairs and household management, which are easily batched. Daily tasks such as cooking and cleaning cannot be postponed in the same way, so the Sunday response in these tasks is a closer guide to what a shorter working day would do.

Second, Sunday synchronises the household. Children are out of school, many wives who work are off, and friends and relatives are free. This raises the value of leisure on Sundays relative to a day off

taken alone, and it also changes the amount of unpaid work to be done. The Saturday comparison and the heterogeneity by children speak to this. A reform that shortened every working day, rather than lengthening the weekend, would release time when the rest of the household is not at home. On these grounds it might move more of the released time into unpaid work.

Third, the estimate is local to compliers, regular salaried men who work on weekdays and not on Sundays. Regular salaried employment accounts for 27.0% of employed men aged 20 to 59 in the sample. The response of casual and self-employed men to a reduction in working time could differ, and the design does not identify their response because Sunday is a less predictable shock to their paid work.

7.2 Measurement

Diaries reported by another household member may understate men's unpaid work (Lee and Waite 2005), and Sunday diaries are 4.6 percentage points (SE 0.9) less likely to be proxy-reported. With proxy reporting controlled for, the unpaid-work share is 15.9 minutes per 100, and among self-reported diaries it is 16.3, against 16.1 in the baseline. The TUS records the main activity in each half-hour slot and, separately, simultaneous activities. A man who minds children while watching television is recorded as doing one of the two as the major activity. If such passive care rises on Sundays and is recorded as leisure, the unpaid-work share understates men's involvement. Secondary activities could be used to bound this.

7.3 Implications for policy

If the exclusion restriction holds, only a small fraction of the time released from paid work goes into unpaid work. On this evidence, shorter working hours alone would do little to change the gender division of unpaid work. The comparison with paternity leave is instructive. Leave policies that reserve time for fathers combine released time with an explicit expectation of care, and some of them have lasting effects on fathers' involvement (Patnaik 2019; Tamm 2019), while others do not (Ekberg et al. 2013). A day off provides time without the expectation. If men use it mostly for rest, the binding constraint on their unpaid work is not time, and policies that work on expectations, through leave reserved for fathers, school and childcare schedules, or information about other men's behaviour (Bursztyn et al. 2020), are more promising than policies that only shorten hours. The wife's response matters as much as the husband's: the narrowing of the within-couple gap on Sundays shows that some of the husband's additional unpaid work relieves the wife.

7.4 Limitations

The exclusion restriction cannot be proven. The falsification test and the difference-in-reduced-forms estimator address the most obvious direct channels, but they rely on men not in paid work being comparable to salaried men in how Sunday affects their unpaid work, and these men are a small, selected group. The diary covers a single day per person, so the comparison is across people

interviewed on different days, not within a person's week (Frazis and Stewart 2012). Wives' time use is observed on the same day as the husband's, but the measure of joint free time is an upper bound, because the analysis panel holds daily totals rather than the timing of episodes. Finally, the survey does not record whether a respondent's employer gives Sunday off, so the first stage is estimated rather than known.

8 Conclusion

The weekly day off is a recurring experiment in men's time. India's Time Use Survey 2024, by recording the day before the interview for households interviewed across the week, turns it into a quasi-random shock to regular salaried men's paid work. These men allocate 16.1 of every 100 minutes released from paid work to unpaid domestic and care work, a statistically significant share, and 58.3 to free time and 22.7 to sleep. Their wives' unpaid work responds enough that the within-couple gap narrows on Sundays. Time availability is not the main constraint on men's unpaid work, and working-time policy alone is unlikely to change the division of unpaid work in India. The same design is available in any time-use survey that records "yesterday" and spreads interviews across the week, and applying it elsewhere would show whether this conclusion is specific to India.

Declaration of generative AI use

Generative AI tools (Claude, Anthropic; and other AI assistants) were used to assist with writing and debugging analysis code, checking references, and drafting and editing text. The author reviewed and verified all analyses, results, and text, and takes full responsibility for the content.

Data and code availability

The Time Use Survey 2024 unit-level data are distributed by the National Statistics Office, Ministry of Statistics and Programme Implementation, Government of India (microdata.gov.in, catalogue 236), to registered users. Code available from the author on request; will be made public on acceptance.

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Appendix

Table 10: Distribution of diary days

Day	All persons 20–59	Regular salaried men 20–59
Mon	14.5%	14.8%
Tue	14.5%	14.7%
Wed	14.3%	14.2%
Thu	14.3%	14.4%
Fri	14.4%	14.8%
Sat	14.3%	14.3%
Sun	13.7%	12.8%

Notes: Survey-weighted share of diaries by the day of the week the diary refers to (the day before the interview).

Table 11: Construction and validation of the person-day panel

Check	Value
Activity records	10,211,478
Person-days (diaries)	450,457
Households	139,487
First Stage Units	9,969
Diaries summing to 1,440 minutes	100.0%
Records flagged minor (secondary)	17.5%
Persons with varying weight	0.00%
Invalid activity codes	None
Married persons with linked spouse	88.5%
Under-6 count method	direct count (file contains members under 6)

Notes: All activity records are used; no row sampling.